JEE

Arjuna JEE 2027

Maths by Amarnath Anand Sir

Quadratic Equations

DPP: 4

Total Duration - 45 Min

Q1 The set of values of p for which the roots of the equation $3x^2+2x+p(p-1)=0$ are of opposite signs is:

(A) $[1,\infty)$
(B) $(-\infty,0]$
(C) $(0,1)$
(D) None of these

Q2 If the roots of $x^2-bx+c=0$ are two consecutive integers, then b^2-4c is

(A) 0
(B) 1
(C) 2
(D) None of these

Q3 The values of k for which the quadratic equation $kx^2+1=kx+3x-11x^2$ has real and equal roots are:

(A) -11,-3
(B) 5, 7
(C) 5, –7
(D) NOT

Q4 The sum of the roots of an equation is 2 and sum of their cube is 98, then form the quadratic equation.

(A) $x^2-2x+15=0$
(B) $x^2+2x-15=0$
(C) $x^2-2x-15=0$
(D) $x^2+2x+15=0$

Q5 The number of real roots of the equation $(x-1)^2+(x-2)^2+(x-3)^2=0$ is

(A) 1
(B) 2
(C) 3
(D) None of these

Q6 If α,β are the roots of the equation $x^2-p(x+1)-c=0$, then $(\alpha+1)(\beta+1)=$

(A) c
(B) $c-1$
(C) $1-c$
(D) None of these

Q7 The quadratic equation with rational coefficient having one root as $1+\sqrt{3}$,is

(A) $x^2+2x+2=0$
(B) $x^2-2x+2=0$
(C) $x^2+2x-2=0$
(D) $x^2-2x-2=0$

Q8 The roots of $4x^2+6px+1=0$ are equal, then the value of p is

(A) $4/5$
(B) $1/3$
(C) $2/3$
(D) $4/3$

Q9 The quadratic equation whose one root is $\frac{1}{2+\sqrt{5}}$ will be

(A) $x^2+4x-1=0$
(B) $x^2+4x+1=0$
(C) $x^2-4x-1=0$
(D) $\sqrt{2}x^2-4x+1=0$

Q10 The roots of the equation $x^2+2\sqrt{3}x+3=0$ are

(A) Real and unequal
(B) Rational and equal
(C) Irrational and equal
(D) Irrational and unequal

Q11 The value of k for which $2x^2-kx+x+8=0$ has equal and real roots are

(A) -9 and -7
(B) 9 and 7
(C) -9 and 7
(D) $9,-7$

Q12 The complete set of values of k, for which the quadratic equation $x^2-kx+k+2=0$, has equal roots consists of

(A) $2+\sqrt{12}$
(B) $2\pm\sqrt{12}$
(C) $2-\sqrt{12}$

(D) $-2-\sqrt{2}$

Q13 If a, b are the roots of the equation $x^2+x+1=0$, then $a^2+b^2=$
(A) 1 (B) 2
(C) -1 (D) 3

Q14 Roots of $\mathrm{a}x^2+\mathrm{b}=0$ are real and distinct, if
(A) $ab>0$
(B) $ab<0$
(C) $a,b>0$
(D) $a,b<0$

Q15 If α and β are the roots of the equation $x^2-4x+1=0$ the value of $\alpha^3+\beta^3$ is
(A) 76 (B) 52
(C) -52 (D) -76

Q16 If the roots of the equation $12x^2-mx+5=0$ are in the ratio $2:3$, then $m=$
(A) $5\sqrt{10}$
(B) $3\sqrt{10}$
(C) $2\sqrt{10}$
(D) $10\sqrt{5}$

Q17 The values of k, so that the equation $2x^2+kx-5=0$ and $x^2-3x-4=0$ may have one root in common.
(A) $-3,-1$
(B) $-3,\frac{-27}{4}$
(C) $-1,-2$
(D) $3,\frac{27}{4}$

Q18 If the equations $ax^2+bx+c=0$ and $x^2+x+1=0$ has one common root then $a:b:c$ is equal to
(A) $1:1:1$
(B) $1:2:3$
(C) $2:3:1$
(D) $3:2:1$

Q19 The roots of the equation $x^2-x-3=0$ are-
(A) Imaginary (B) Rational
(C) Irrational (D) None of these

Q20 If the roots of the equation $x^2-10x+21=m$ are equal then m is-
(A) 4 (B) 25
(C) –4 (D) 0

Answer Key

Q1	(C)	Q11	(D)
Q2	(B)	Q12	(B)
Q3	(C)	Q13	(C)
Q4	(C)	Q14	(B)
Q5	(D)	Q15	(B)
Q6	(C)	Q16	(A)
Q7	(D)	Q17	(B)
Q8	(C)	Q18	(A)
Q9	(A)	Q19	(C)
Q10	(C)	Q20	(C)